Design Document for MusiBox

Group 3_Swarna_1

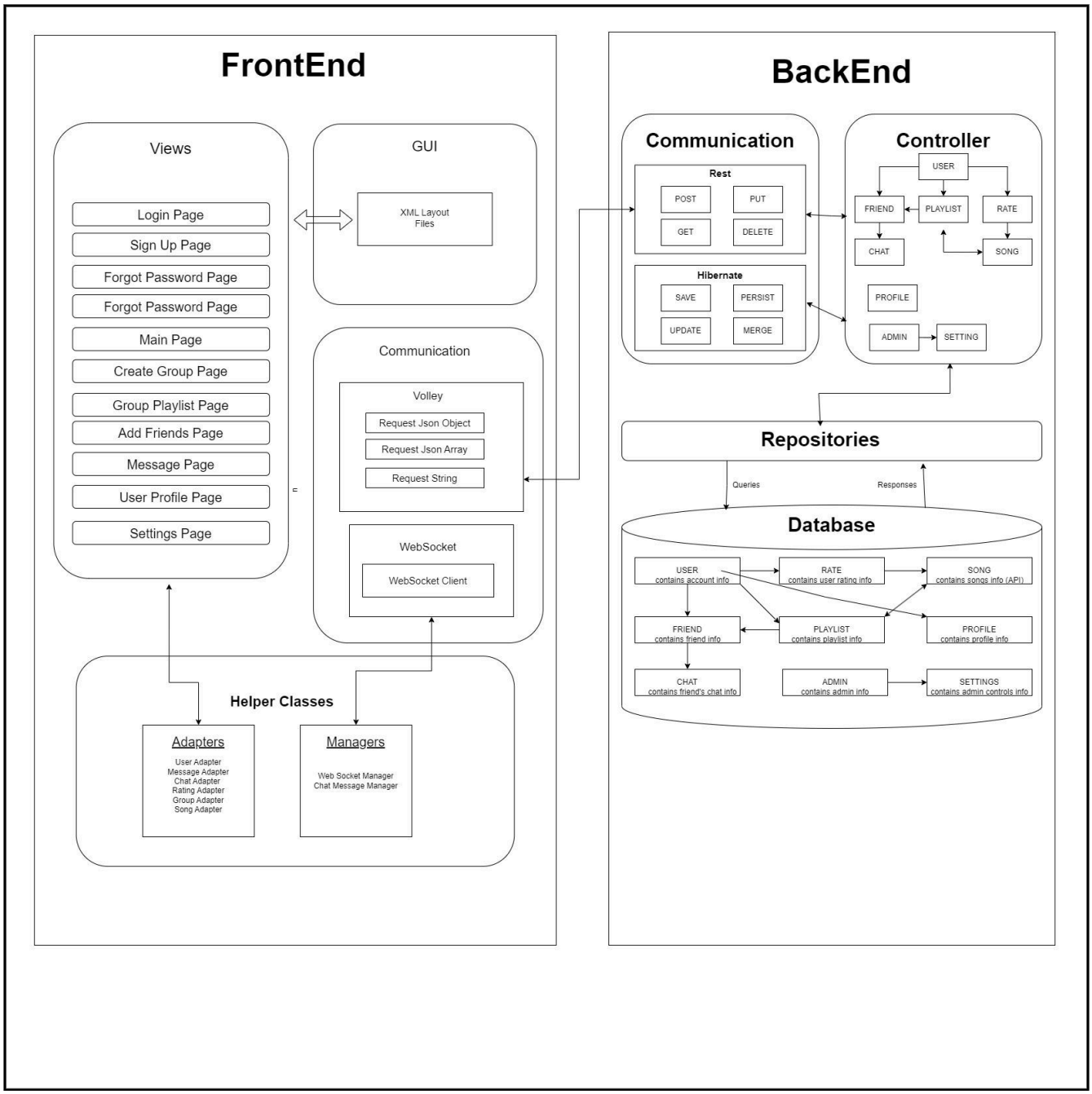
Tu Reh: 25% contribution

Ping Wu: 25% contribution

Bae Meh: 25% contribution

Haris Khan: 25% contribution

Page 1. Block Diagram



Frontend (*Main Implementations)*

Main Page

- A page where users can rate songs live by either scrolling or using a search bar and see the live ratings update:
    - EditText : Search bar Field
    - Recycler View: Rating bar
    - Recycler View: Live Avg Ratings
    - Image Button: Bottom Navigation bar

Group Creation

- The page where users can create a group by selecting users, using the search bar and then the groups created will then be clickable to bring users to the group playlist page:
    - EditText : Search Bar
    - ChipGroup: Shows the users selected
    - Button: Create group button
    - Recycler View: List of Groups created
    - Image button: Bottom Navigation Bar

Playlist

- A Shared Playlist between multiple users.On this page the users are able to see what songs have been added to the playlist and are able to remove songs.
    - Recycler View: Playlist songs
    - TextView: Group name
    - Image Button: Back button to group creation
    - Button: Delete song

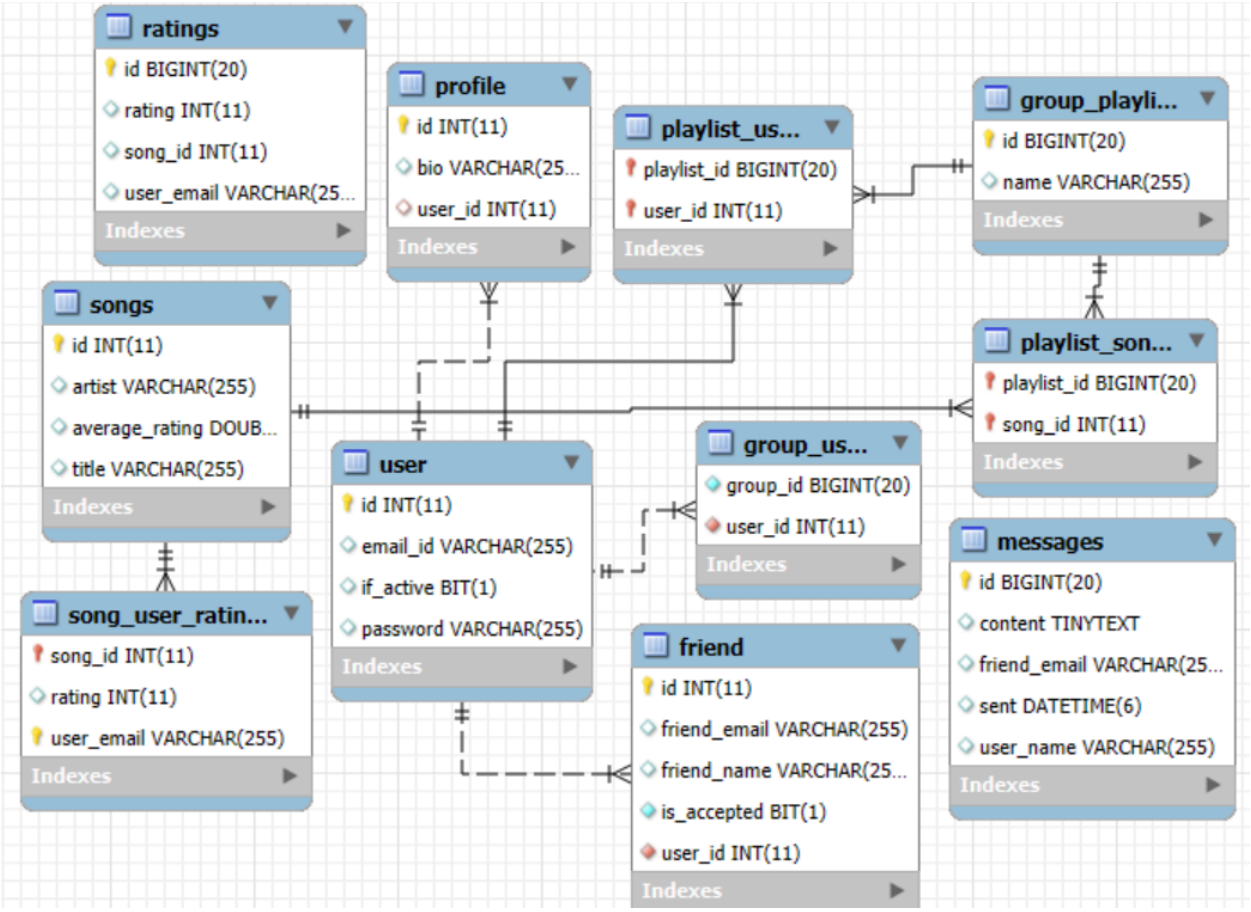
Backend

The backend updates the database based on information sent to the given mappings. These include:

- Post: send information on an item to be added to the database.
- Get: request information, often with an identifier for the specific item requested from the database
- Put: send information to update a specific item in the database
- Delete: send an identifier to delete a specific item from the database

Controllers: The controllers contain the endpoints which mappings for communication between frontend and the database:

- User: Contains the above mappings to create users, which contain one-to-one relationships with the below profile, one to many with friends, and one to many with groups.
- Friend: many to one relationship with user.
- Profile: a user's profile which contains their bio.
- Songs: contains the title and artist and the average rating. Has a many to many relation with group playlists.
- Song_user_ratings: contains the ratings for songs each user posted, if user has multiple ratings only the latest one is valid.
- ratings: Same as song_user ratings with a one to many relationship,
- Playlist_users: Contains the users in the group playlist which can be added and removed, many to many with users.
- Group_playlist: Contains the name for the group which is connected to the playlist users.
- Playlist_songs: contains the songs inside each playlist.
- Messages: Stores the messages sent between two users which are friends.


ratings
id BIGINT(20)
rating INT(11)
song_id INT(11)
user_email VARCHAR(25...
Indexes
profile
id INT(11)
bio VARCHAR(25...
user_id INT(11)
Indexes
playlist_us...
playlist_id BIGINT(20)
user_id INT(11)
Indexes
group_playli...
id BIGINT(20)
name VARCHAR(255)
Indexes
songs
id INT(11)
artist VARCHAR(255)
average_rating DOUB...
title VARCHAR(255)
Indexes
playlist_son...
playlist_id BIGINT(20)
song_id INT(11)
Indexes
user
id INT(11)
email_id VARCHAR(255)
if_active BIT(1)
password VARCHAR(255)
Indexes
group_us...
group_id BIGINT(20)
user_id INT(11)
Indexes
messages
id BIGINT(20)
content TINYTEXT
friend_email VARCHAR(25...
sent DATETIME(6)
user_name VARCHAR(255)
Indexes
song_user_ratin...
song_id INT(11)
rating INT(11)
user_email VARCHAR(255)
Indexes
friend
id INT(11)
friend_email VARCHAR(255)
friend_name VARCHAR(25...
is_accepted BIT(1)
user_id INT(11)
Indexes